From clinical notes to clinical screening:
Using AI to turn routine health records into actionable evidence

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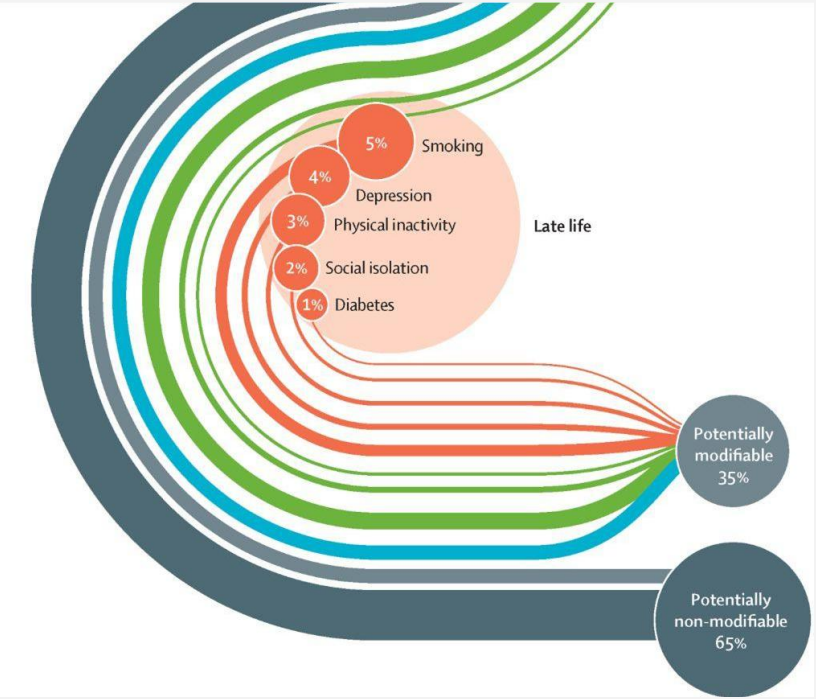
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Loneliness and Social isolation in Dementia

Between 2 to 5% dementia cases attributable to social isolation Livingston, et al. (2024)

2025 meta-analysis estimated loneliness in about 42.7% of people with dementia, while evidence on social isolation in cognitive impairment was even higher (~60%) Hajek & König, (2025).

No routine assessment within primary or secondary practice



Mental health information: clinical note

- Tabulated data that can be used for research
- 80% of clinical information in rich narrative form

I reviewed XXXX on 1 December 2015.
XXXX is a 77-year-old woman with a history of depression and a diagnosis of Alzheimer's dementia.

Mental state examination
XXXX appeared well kept and good rapport was established. Her mood was subjectively anxious. She reported that she feels increasingly anxious as the evening approaches due to loneliness and often phones her daughter for support.

Objectively, she appeared to have some low mood with anxiety.
MMSE: 23/30

Various symptoms noticed and documented as part of the clinical assessment

But the information of interest was locked in free-text clinical notes

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How do we find and scale this across million of records?

Electronic Health Records (EHRs)

Data are de-identified patient records that cover each interaction and intervention between a patient and clinicians

Mentions of loneliness and social isolation in the text:

- “XXXX lives alone in her own home”
- “social isolation, lives on the 2nd floor flat with no lift, can currently manage stairs”
- “she said she feels increasingly anxious as the evening approaches due to loneliness so she end up phoning her daughter”
- “He is desperately lonely and has diagnosis of Alzheimer’s”

Language model



Accuracy

"Wife died in April of this year – he feels lonely but is coping well"

Loneliness reports — 0.91

"has been sitting in the tv room for a while…has isolated herself from other patients"

Non-informative isolation

"Concerned about leaving as she feels socially isolated"

Social isolation reports — 0.83

"Had a lovely day with family"

Non-informative sentence

Human-labelled examples → Language model → Classification

We let it read

6,388,715 clinical documents from 34,469 patients

What we see in the data :

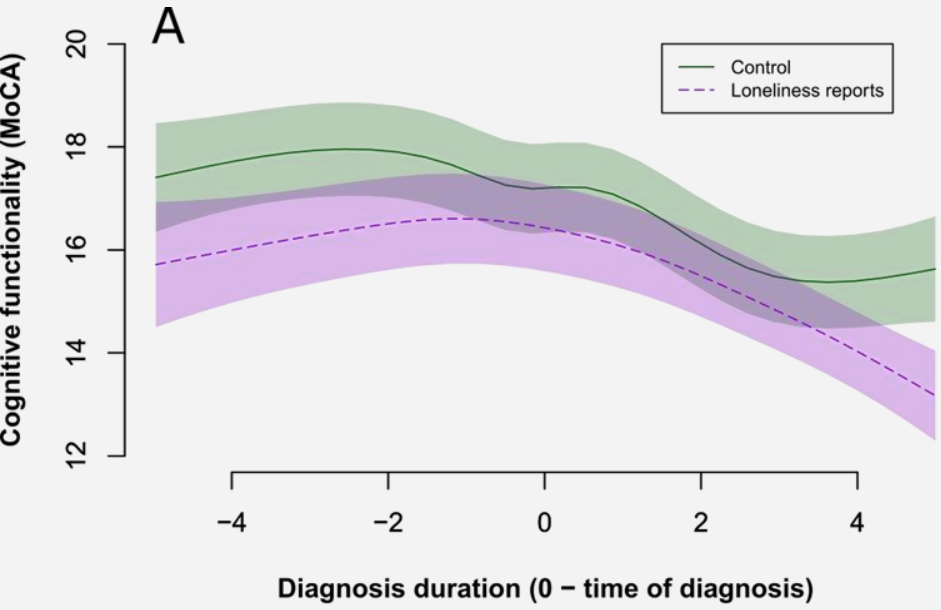
- Social isolation reported in 19% of the cases
- Loneliness reported in 14%
- Most commonly documented in nursing and occupational therapy notes

Are these reports related to clinical outcomes?

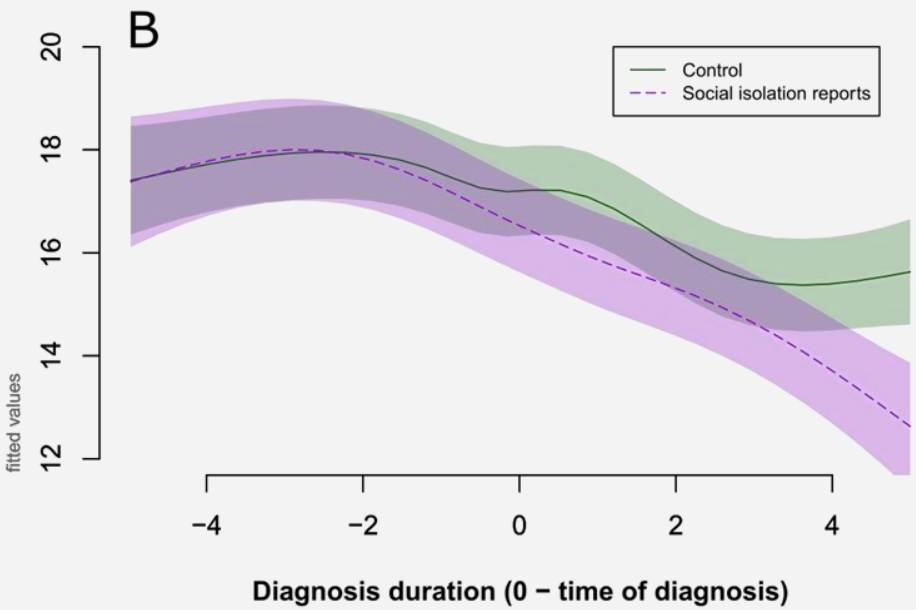


Cognitive trajectories

Loneliness: lower cognition across the trajectory



Social isolation: stronger decline before diagnosis



Translation into practice:

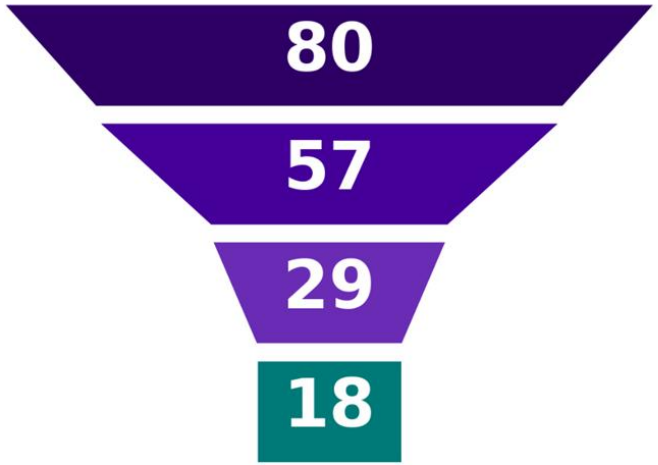
Together with Working-age memory clinic at Sheffield Teaching Hospitals

- Reviewed psychometric tools for Social isolation and loneliness
- Screen people with dementia during clinical appointments using:
  - UCLA-3 item loneliness scale (< 5 minutes) and Lubben Network Scale (5-10 minutes)
- Move away from inferring social need to asking about it directly and measuring the gap

Results from clinic:

From clinic to social prescribing: the referral gap

Working-age memory clinic service evaluation · n = 80 patients



Seen in clinic
100% of attendees

Completed screening
71% · UCLA-3 + Lubben

Significant social burden
high loneliness or isolation

Referred to social prescribing
of the 29 with burden

11 **patients with social burden were NOT referred**

38% of those who screened positive — a missed opportunity at the point of care

Conclusions

Clinical text contains meaningful information beyond diagnoses and treatments: NLP allows this information to be examined systematically and at scale.

EHRs can help identify gaps in how social needs are recorded and addressed: Loneliness and social isolation were present in clinical notes, but routine documentation is unlikely to capture their full burden.

Systematic screening provides a clearer picture of need: 38% of patients screened positive for loneliness or social isolation, and around one third of those with identified need were not referred for support.

NLP can help us understand where additional assessment and support may be useful, but identifying need is only the first step in improving care.

Thank you for your attention



University of Sheffield

Thank you to the team:
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